ls
1
grep
1
true
2
nl
3
nl
4
nl
5
grep
6
grep
6
7
nl
8
nl
9
python3
10
nl
11
python3
12
git
13
python3
14
python3
15
echo
15
git
15
git
localization
patch
validation
general

```
.../n", outer)print("shapes:", m.shape, mt.shape, hnf.shape, hnf_t.shape)
```

```
.../from sympy import matrixfrom sympy.matrices.normalforms import hermite_normal_formarr = [[5,8,12],[0,0,1]]# simulate np.flip (reverse both axes)def flip(a):    return [list(reversed(row)) for row in reversed(a)]inner = flip(arr)print("inner (flipped):", inner)m = matrix(inner)print("m:", m)mt = m.tprint("mt:", mt)hnf = hermite_normal_form(mt)print("hnf:", hnf)hnft = hnf.tprint("hnf.t:", hnft)outer = flip([list(row) for row in hnft.tolist()])print("outer (flipped hnf.t):", outer)
```

```
.../normalforms.py (backup at {})'.format(bak))
```

```
.../from sympy import matrixfrom sympy.matrices.normalforms import hermite_normal_formarr = [[5,8,12],[0,0,1]]def flip(a):    return [list(reversed(row)) for row in reversed(a)]inner = flip(arr)print("inner (flipped):", inner)m = matrix(inner)print("m:", m)mt = m.tprint("mt:", mt)hnf = hermite_normal_form(mt)print("hnf:", hnf)hnft = hnf.tprint("hnf.t:", hnft)outer = flip([list(row) for row in hnft.tolist()])print("outer (flipped hnf.t):", outer)
```